CS 374: Section A - Homework 8, Problem 2

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The label of the first vertex v_0 of a walk is not part of $\sigma(W)$. So when a walk moves along an edge $u \to v$, it reads exactly one character, the label $\ell(v)$ of the vertex it arrives at.

The idea in both parts is the same: run a DFA alongside the walk. A vertex of the new graph records the current station in G and the current state of the DFA, and each edge moves in G and feeds the new label to the DFA.

(a) Walks whose string contains 374

DFA

The DFA M_{374} has states $0, 1, 2, 3$. State j means "the longest suffix of the string read so far that is a prefix of 374 has length j," and state 3 means 374 has already appeared. It starts in state 0, and state 3 is the only accepting state. Its transitions $\delta(j, d)$ are:

state	3	7	4	other digits
0	1	0	0	0
1	1	2	0	0
2	1	0	3	0
3	3	3	3	3

For example, from state 2 (the string ends in 37) the digit 3 goes to state 1, because the string now ends in 373, whose longest suffix that is a prefix of 374 is 3. The DFA ends in state 3 exactly when its input contains 374.

Graph

Build a directed graph $H = (V_H, E_H)$:

- **Vertices:** $V_H = V \times \{0, 1, 2, 3\}$. Vertex (v, j) means "the walk is at vertex v, and M_{374} is in state j after reading the string of the walk so far." So $|V_H| = 4\,|V|$.
- **Edges:** For every edge $u \to v$ in E and every state j, H has an edge $(u, j) \to (v, \delta(j, \ell(v)))$. So $|E_H| = 4\,|E|$.

Algorithm

> Run whatever-first search (for example, BFS) in H from $(s, 0)$.
> Return TRUE if $(t, 3)$ is reached, and FALSE otherwise.

Correctness

Claim: The vertex (v, j) is reachable from $(s, 0)$ in H if and only if G has a walk W from s to v such that M_{374} ends in state j after reading $\sigma(W)$.

Walks give paths: We use induction on the number of edges of W. A walk with no edges has $v = s$ and $\sigma(W) = \varepsilon$, so $j = 0$, and $(s, 0)$ is reachable from itself. Otherwise W is a walk W' from s to some u, followed by the edge $u \to v$, and $\sigma(W) = \sigma(W')\ell(v)$. If M_{374} ends in state i on $\sigma(W')$, then (u, i) is reachable by induction, and M_{374} ends in state $j = \delta(i, \ell(v))$ on $\sigma(W)$. The edge $(u, i) \to (v, j)$ is in H, so (v, j) is reachable.

Paths give walks: We use induction on the number of edges of a path from $(s, 0)$. The empty path ends at $(s, 0)$, which matches the walk $\langle s \rangle$. Otherwise the last edge is $(u, i) \to (v, j)$, coming from an edge $u \to v$ of G with $j = \delta(i, \ell(v))$. By induction, there is a walk W' from s to u on whose string M_{374} ends in state i. Adding the edge $u \to v$ gives a walk from s to v on whose string M_{374} ends in state j.

By the claim with $v = t$ and $j = 3$, the search reaches $(t, 3)$ if and only if some walk W from s to t drives M_{374} to state 3, that is, if and only if $\sigma(W)$ contains 374.

Running time

Building H takes $O(|V_H| + |E_H|) = O(|V| + |E|)$ time, since each transition is a table lookup. The search also takes $O(|V_H| + |E_H|)$ time. The algorithm runs in

$$\boxed{O(|V| + |E|) \text{ time.}}$$

Sources and collaborators: None.

LLM use: I used Claude (Anthropic) as a tutor for this problem: it helped me understand the problem and guided me, through explanation and questions, toward a correct approach. I wrote and typeset this entire submission myself.

(b) Walks whose string is accepted by a DFA M

Let $M = (Q, \delta, s', A)$, and let $\delta^*(q, x)$ be the state M reaches from state q after reading the string x.

Graph

Build a directed graph $H = (V_H, E_H)$, the product of G and M:

- **Vertices:** $V_H = V \times Q$. Vertex (v, q) means "the walk is at vertex v, and M is in state q after reading the string of the walk so far." So $|V_H| = |V|\ |Q|$.
- **Edges:** For every edge $u \to v$ in E and every state $q \in Q$, H has an edge $(u, q) \to (v, \delta(q, \ell(v)))$. So $|E_H| = |E|\ |Q|$.

Algorithm

> Run whatever-first search (for example, BFS) in H from (s, s').
> Return TRUE if some vertex (t, q) with $q \in A$ is reached, and FALSE otherwise.

After the search, we check the $|Q|$ vertices (t, q) one by one.

Correctness

Claim: For every vertex $v \in V$ and state $q \in Q$, the vertex (v, q) is reachable from (s, s') in H if and only if G has a walk W from s to v with $\delta^*(s', \sigma(W)) = q$.

Walks give paths: We use induction on the number of edges of W. If W has no edges, then $v = s$, $\sigma(W) = \varepsilon$ and $q = s'$, and (s, s') is reachable from itself. Otherwise W is a walk W' from s to some vertex u, followed by the edge $u \to v$, and $\sigma(W) = \sigma(W')\ell(v)$. Let $p = \delta^*(s', \sigma(W'))$. By induction, (u, p) is reachable. Then $\delta^*(s', \sigma(W)) = \delta(p, \ell(v)) = q$, so the edge $(u, p) \to (v, q)$ is in H, and (v, q) is reachable.

Paths give walks: We use induction on the number of edges of a path from (s, s') to (v, q). A path with no edges ends at (s, s'), and the walk $\langle s \rangle$ has $\sigma = \varepsilon$ and $\delta^*(s', \varepsilon) = s'$. Otherwise the last edge of the path is $(u, p) \to (v, q)$ for some edge $u \to v$ in E with $q = \delta(p, \ell(v))$. By induction, G has a walk W' from s to u with $\delta^*(s', \sigma(W')) = p$. Then W' followed by the edge $u \to v$ is a walk W from s to v with $\delta^*(s', \sigma(W)) = \delta(p, \ell(v)) = q$.

Conclusion: By the claim, the search reaches some (t, q) with $q \in A$ if and only if G has a walk W from s to t with $\delta^*(s', \sigma(W)) \in A$, that is, with $\sigma(W) \in L(M)$. Whatever-first search reaches exactly the vertices reachable from its start, so the algorithm is correct. This also covers the walk with no edges: if $s = t$ and $s' \in A$, the answer is TRUE because $\varepsilon \in L(M)$.

Running time

Building H takes $O(|V_H| + |E_H|) = O((|V| + |E|)\ |Q|)$ time, since each transition $\delta(q, \ell(v))$ takes $O(1)$ time. The search takes $O(|V_H| + |E_H|)$ time, and checking the vertices (t, q) takes $O(|Q|)$ time. The algorithm runs in

$$\boxed{O((|V| + |E|)\ |Q|) \text{ time.}}$$

With $|Q| = 4$ this is $O(|V| + |E|)$, matching part (a).

Sources and collaborators: None.

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